

A Project On AYH MART - Self Checkout System Inclusive Integrated | Innovative

Programme
IMCA I Sem-2

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What we'll cover

Agenda

- 1 . Project Abstract
- 2 . Objective
- 3 . System Description
- 4 . Screenshots
- 5 . Test Cases
- 6 . References

Project abstract

AYH Mart self checkout system

AYH Mart is a console-based self checkout system developed in C, designed for a retail store in Surat, India. It allows customers to independently browse products, add items to a cart, and complete purchases via Cash or UPI QR Code payment.

The system supports three user roles - Customer, Staff, and Admin - each with a dedicated portal. Key features include product inventory management, a tiered membership discount program (Silver, Gold, Platinum), customer account management, and a detailed sales reporting module.

All data is managed in-memory using C structures, making the system lightweight, efficient, and fully functional as a retail management solution.



Project objectives



C-based system

Functional self-checkout system built in C for a real retail environment.



Role-based access

Separate access levels for Admin, Staff, and Customer users.



Inventory management

Add, Edit, Delete, and View products with cart and stock deduction.



Tiered membership

Silver (5%), Gold (10%), and Platinum (15%) discount tiers for members.



Dual payment

Cash with change calculation and UPI QR Code payment support.



Sales reports

Revenue, discount, and product-wise reports with structured data records.

System description

Languages, libraries, data structures and architecture

Language & Platform

- C (Standard C)
- Windows Console App
- GCC / MinGW / Dev-C++ / Code::Blocks

Libraries used

- `stdio.h` — I/O operations
- `string.h` — String handling
- `ctype.h` — Char classification
- `stdlib.h` — System calls, `exit()`
- `windows.h` — UTF-8 / Unicode support

Architecture

Menu-driven, modular function design.

5 core modules:

- Inventory
- Members
- Customer Self-Checkout
- Sales Reports
- Customer Accounts

Data structures (structs)

Product

`pCode · pName · pPrice`
`· pStock · pCategory`

Member

`mId · mName · mSpent ·`
`mTier · mDisc · mUsed`

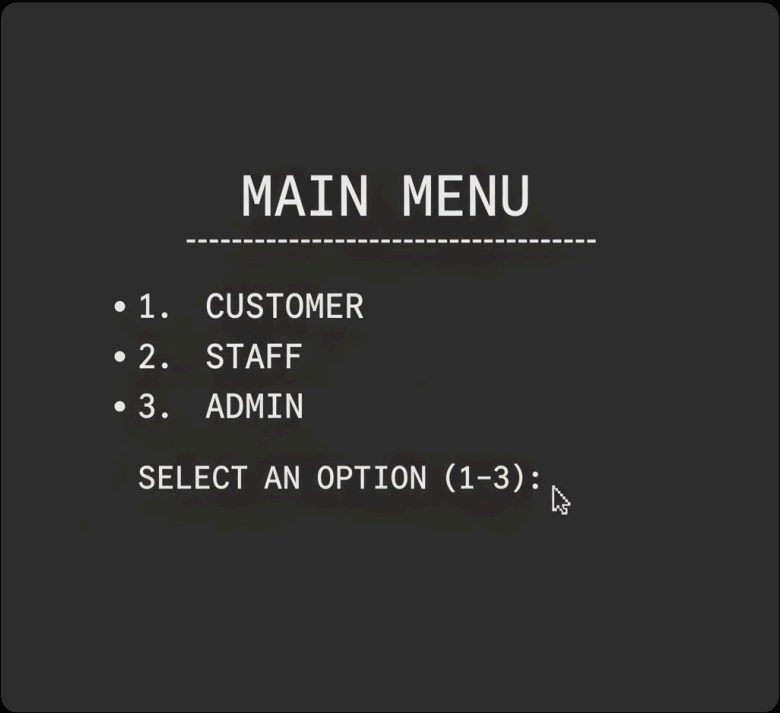
CartItem

`code · name · price · qty`

Sale & Customer

Sale: `pCode · pName · qty ·`
`price · total`
Customer: `cId · cName · cPass ·`
`cMemberId`

Screenshots



Main menu and login

- Role selection: Customer, Staff, Admin
- Customer login and self-checkout menu



Product and cart flow

- Product list: Code, Name, Price, Stock
- Add to cart and Admin inventory table



Checkout and payment

- Subtotal, discount, payable amount
- QR code UPI payment and sales report

Test cases

All 12 test cases passed. AYH MART Self Checkout System - functional validation results.

#	Test Case	Input	Expected Output	Result
01	Customer login - valid	Correct ID and password	"Welcome, Name!"	Pass
02	Customer login - invalid	Wrong ID or password	Invalid Customer ID or Password	Pass
03	Add to cart - valid product	Correct code, qty ≤ stock	Item added to basket	Pass
04	Add to cart - insufficient stock	qty > available stock	Insufficient stock warning	Pass
05	Membership discount - valid	Member ID with active card	Discount applied, card marked Used	Pass

#	Test Case	Input	Expected Output	Result
06	Membership discount - used card	Member ID with used card	Card already used. No discount.	Pass
07	Cash payment - sufficient	Amount $\geq$ payable	Change returned, bill printed	Pass
08	Cash payment - insufficient	Amount $<$ payable	Need X more prompt	Pass
09	QR Code payment	Select QR option	QR displayed, Transaction ID generated	Pass
10	Admin delete product - valid	Valid product code	Product deleted successfully	Pass
11	Admin delete product - invalid	Invalid product code	Product code not found	Pass
12	Sales summary report	After 2 transactions	Correct bills, revenue, discounts shown	Pass

Refer- ences

Books

Kernighan & Ritchie — The C Programming Language, 2nd Ed.,
Prentice Hall

Reema Thareja — Data Structures Using C, Oxford University Press

Online resources

Microsoft Docs — Windows Console API and SetConsoleOutputCP
documentation

GeeksforGeeks — C structs, string operations, and menu-driven
programs ([geeksforgeeks.org](https://www.geeksforgeeks.org))

Lectures

NPTEL — Programming in C, IIT Bombay

Thank You

AYH MART
Self Checkout System